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WEB SCRAPING USING PYTHON

A beginner-friendly session with
Beautiful Soup

DEVELOPERS CLUB
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Why Use Web Scraping?

- Web scraping is useful when:
 - The site has no public API.
 - The API is limited or rate-restricted.
 - You need data from multiple pages or categories.
- Not all websites offer structured access to data through APIs.



How Web Scraping Works ?

- Web scraping involves two main components:

CRAWLER	SCRAPER
A crawler finds where to go	A scraper extracts what to get
A crawler (also called a spider or bot) is a program that automatically navigates through websites, following links from page to page.	A scraper is a program that extracts specific data from a web page, like titles, prices, images, reviews, etc.
Can be built using loops + requests to follow links (href)	Often built using BeautifulSoup or Selenium



Essential Packages and Tools for Python Web Scraping

1. Requests - Fetch HTML pages from internet
2. BeautifulSoup – Parse HTML/XML content
3. lxml- Fast HTML/XML content
4. Selenium – Automate Browser actions
5. Scrapy – Full-fledged crawling & scraping framework
6. urllib – Basic URL handling
7. Pandas – Clean and Organise scraped data



Requests – Fetching Web Pages

What is requests ?

- A popular python library used to send HTTP requests to websites and get data from them.
- Used to fetch the raw HTML content from web pages before we pass it to BeautifulSoup for parsing.

USAGE :

```
import requests

url = 'https://example.com'
response = requests.get(url)
print(response.status_code)    # 200 means OK
print(response.text)          # HTML content
```

Common Status Code

CODE	MEANING
200	OK
301/302	Redirect
403	Forbidden
404	Not Found
500	Server Error



BeautifulSoup

- ❑ Python package for parsing HTML & XML docs.
- ❑ Helps in navigating, searching and extracting data from web pages.

Use Cases :

- ❑ Scraping blog articles, titles, authors.
- ❑ Extracting prices, reviews from e-commerce pages.
- ❑ Collecting headlines from news websites.
- ❑ Pulling tables or stats from Wikipedia

It is a library that allows you to gather any info you want from any website you want

Parsing – Breaking down and analyzing a big chunk of data (like a HTML page) and extract the parts you need.

Think it of like reading a messy website page and picking out only the useful info.

HISTORY - Named after a poem in “Alice’s Adventures in Wonderland”

Technical Meaning

Messy Webpage
(Ugly HTML Soup)

After parsing

Beautiful Soup

Purpose : Helps extract the good stuff from messy HTML

Basic HTML Structure , HTML Tags

- ❑ <h1> to <h6> – Headings (h1 is the largest)
- ❑ <p> – Paragraph (used for blocks of text)
- ❑ – Anchor tag (creates hyperlinks)
- ❑ – Image tag (embeds an image)
- ❑ , , – Lists (unordered/ordered with list items)
- ❑
 – Line break (moves content to the next line)
- ❑ <div> – Block-level container (used with class/id for layout or styling)

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Webpage</title>
  </head>
  <body>
    <div class="container">
      <h1>Hello, World!</h1>
      <p>This is a paragraph inside a
div.</p>
    </div>
  </body>
</html>
```

HTML Parsers

Parsers	Description	Speed / Accuracy	Installation	Best For
html.parser	Built-in Python parser	Medium / Decent	No (Default)	Simple , well-formatted HTML
lxml	Fast and powerful HTML/XML parser	Fast / Very good	pip install lxml	Large files, speed-critical apps
html5lib	Parsers like a browser , very lenient	Slow / Perfect	pip install html5lib	Messy or broken HTML



5 Steps to Web Scraping

01

**Import
Required
Libraries**

02

**Send a get
Request**

03

**Check
response
status
(200 – OK)**

04

**Parse HTML
content**

05

**Prettify and
print HTML**



Q & A

**Any Questions or thoughts
?**



RESOURCES

BeautifulSoup :

- ❑ Official docs : <https://www.crummy.com/software/BeautifulSoup/bs4/doc/>
- ❑ Tutorial (Crash Course) : <https://www.youtube.com/watch?v=ng2o98k983k>
- ❑ Quick Guide : <https://realpython.com/beautiful-soup-web-scraper-python/>

Requests:

- ❑ Official docs : <https://docs.python-requests.org/en/latest/>
- ❑ Quickstart Guide: <https://requests.readthedocs.io/en/latest/user/quickstart/>
- ❑ Article (Real Python): <https://realpython.com/python-requests/>



THANK YOU

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